HACKSAGON

A NATIONAL LEVEL SOFTWARE & HARDWARE HACKATHON

ABV IIITM STUDENT BRANCH

IEEE MP SECTION

संकेतEdge

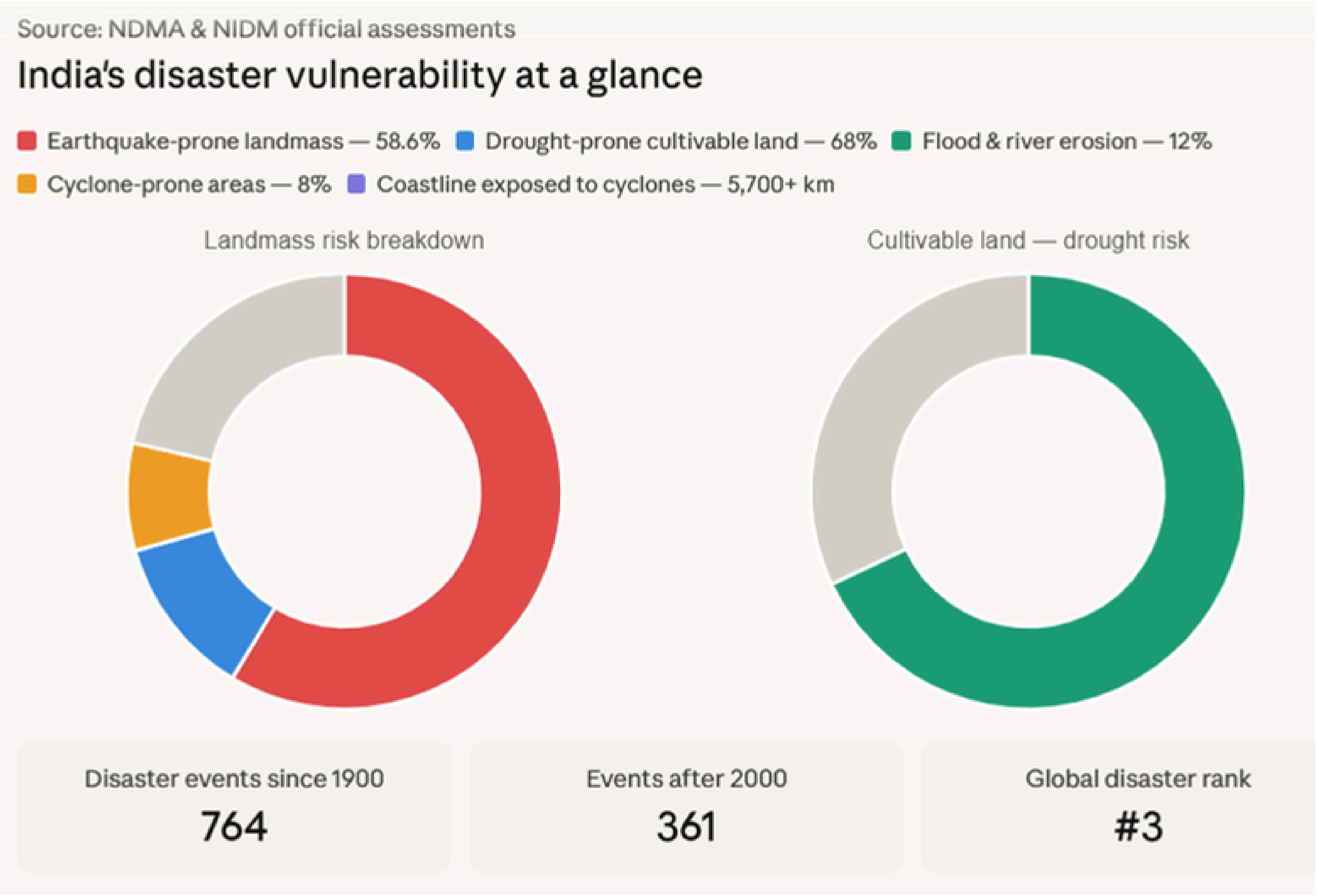
By : Naive Birds

TRACK: SOFTWARE+HARDWARE

THEME: DISASTER MANAGEMENT AND EMERGENCY RESPONSES

INSTITUE: SYMBIOSIS INSTITUTE OF TECHNOLOGY, NAGPUR

WHY ARE DISASTERS STILL SO DESTRUCTIVE?



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THE TIMES OF INDIA

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When disasters strike, preparation saves lives — is India ready when crisis hits?

Oshin Bhatia / TIMESOFINDIA.COM / Updated: Jan 11, 2026, 14:47 IST

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ANI Photo

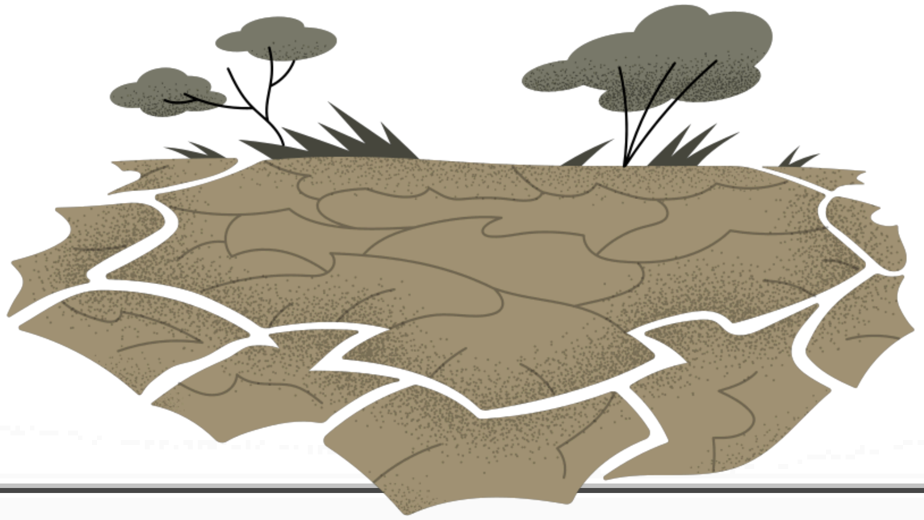
Disasters don't just cause infrastructure losses, crip economies, or claim lives. They shake something far deeper, that is, the very fabric of human existence.

Each disaster leaves scars that statistics cannot full capture silent grief, disrupted childhoods, fractured communities and lives permanently divided into "before" and "after". Homes swallowed by raging flo erase decades of memory. Landslides bury entir villages in a single night. The suffering often s long after the debris is cleared and relief down.

Source : The Times of India (Dated : Jan 11, 2026)

PROBLEM STATEMENT

To design and develop a **low-cost**, autonomous **edge AI-enabled IoT system** that can locally sense and **analyze flood, drought, and heavy rainfall conditions**, generate alerts over **long distances** to at-risk communities, and trigger on-site warning mechanisms without relying on any external network or cloud connectivity



PROPOSED SOLUTION

This study provided the solution for Disaster Management with the help of IOT Hardware and AIML Analytics

Technologies used :

Edge Sensor Nodes

Real-time monitoring of water, soil, temp, humidity with on-device AI risk checks via LoRa.

LORA Technology

Long-range, offline JSON messaging for nodes, no internet needed.

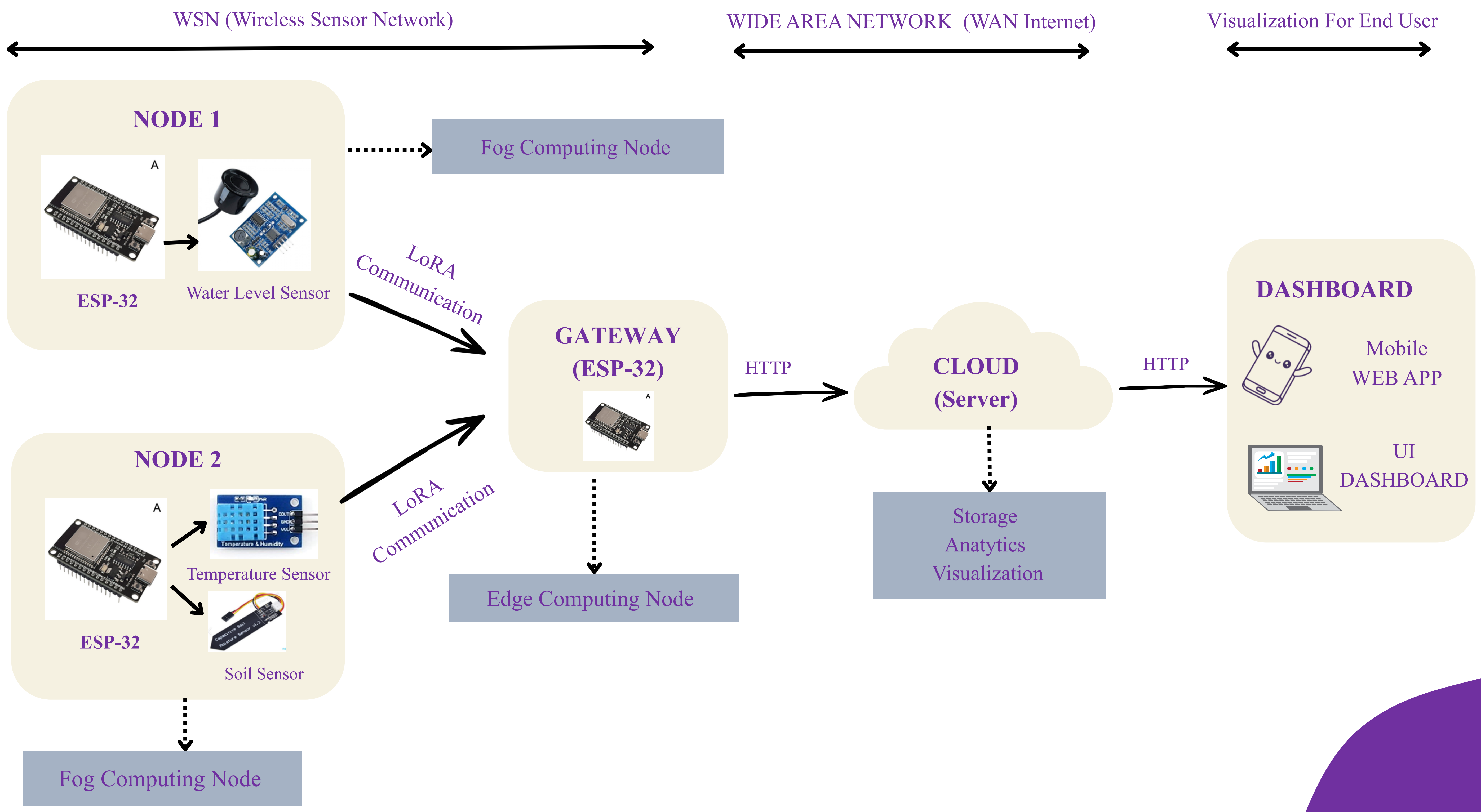
Decision Gateway

Fuses node data, computes flood/drought tiers, triggers on-site alerts.

Cloud & Dashboard

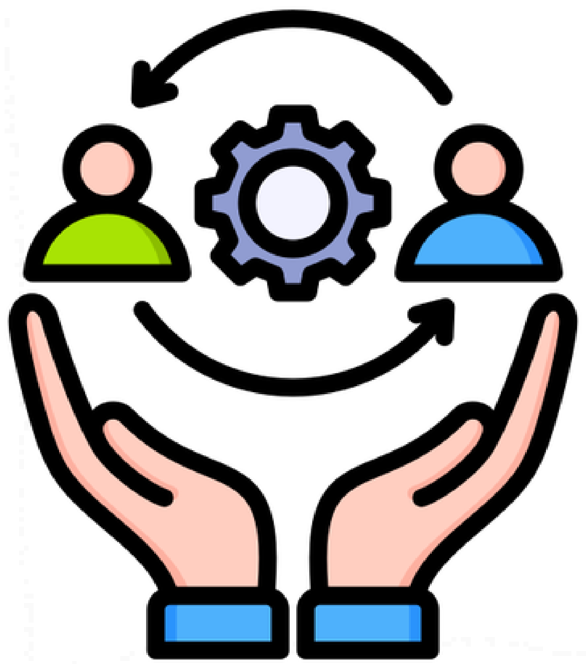
Uploads summaries online; shows live status + alert explanations.

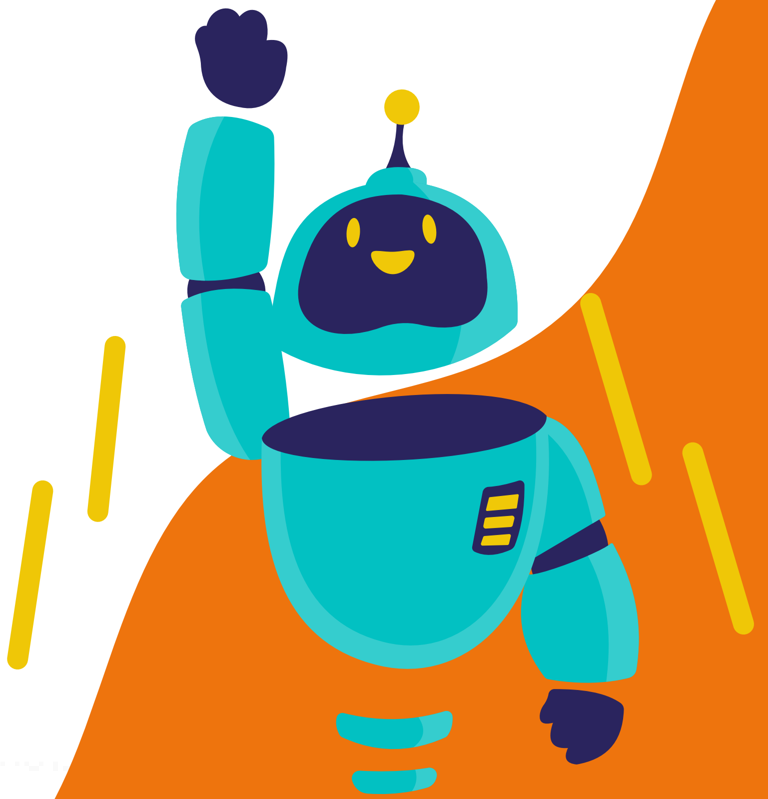
SYSTEM ARCHITECTURE



NOVEL APPROACHES

- Multi-Hazardous System
- Edge Computing and Fog Computing (Using Tiny ML)
- Common Format using JSN
- Explainable AI (SHAP)





FUTURE SCOPE

Backup edge prediction maintains risk assessment when sensors fail.

Scalable system adaptable to multiple disasters type.

Live Dashboard with Real-time mapping for monitoring and decision-making.

CHALLENGES FACED

LoRa module lacked support for the 865MHz band, requiring frequency, bandwidth and adjustment according to devices.

Limited and imbalanced drought dataset required careful preprocessing and handling for effective model training

Multiple LoRa devices on the same frequency caused interference, so frequency has to be adjusted

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THANK YOU